

Just as with the example of catching balls, we see in this buffet example that algorithms like gradient ascent are natural to human life and decision-making. They are natural to us even if we have never taken a math class or written a line of code. The tools in this chapter are merely meant to formalize and make precise the intuitions you already have.

When Not to Use an Algorithm

Often, learning an algorithm fills us with a feeling of power. We feel that if we are ever in a situation that requires maximization or minimization, we should immediately apply gradient ascent or descent and implicitly trust whatever results we find. However, sometimes more important than knowing an algorithm is knowing when not to use it, when it's inappropriate or insufficient for the task at hand, or when there is something better that we should try instead.

When should we use gradient ascent (and descent), and when should we not? Gradient ascent works well if we start with the right ingredients:

- A mathematical function to maximize
- Knowledge of where we currently are
- An unequivocal goal to maximize the function
- Ability to alter where we are

There are many situations in which one or more of these ingredients is missing. In the case of setting taxation rates, we used a hypothetical function relating tax rates to revenue. However, there's no consensus among economists about what that relationship is and what functional form it takes. So we can perform gradient ascent and descent all we like, but until we can all agree on what function we need to maximize, we cannot rely on the results we find.

In other situations, we may find that gradient ascent isn't very useful because we don't have the ability to take action to optimize our situation. For example, suppose that we derived an equation relating a person's height to their happiness. Maybe this function expresses how people who are too tall suffer because they cannot get comfortable on airplanes, and people